AP CALCULUS AB

Unit 1 • Lesson 1.15 practice • Answer key

Lesson 1.15 • Connecting Limits at Infinity and Horizontal Asymptotes

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x\to\infty} \dfrac{3x^2+1}{x^2-5}$ is

(A) 3

(B) $-\dfrac{1}{5}$

(C) 0

(D) ∞

Solution

Answer: (A). Equal degrees: $\frac{3}{1} = 3$.

Question 2

NO CALCULATOR

$\lim_{x\to\infty} \dfrac{x+1}{x^2}$ is

(A) 1

(B) 0

(C) ∞

(D) $\dfrac{1}{2}$

Solution

Answer: (B). The degree of the denominator is larger.

Question 3

NO CALCULATOR

$\lim_{x\to\infty} e^{-x}$ is

(A) 1

(B) ∞

(C) 0

(D) $-\infty$

Solution
Answer: (C). $e^{-x} = \frac{1}{e^x}$ and $e^x \to \infty$.

Question 4

NO CALCULATOR

$\lim_{x\to\infty} \frac{x^2}{e^x}$ is

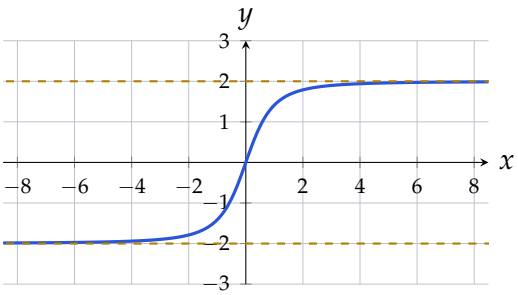
(A) ∞
(B) 1
(C) e
(D) 0

Solution
Answer: (D). Exponential growth dominates polynomial growth.

Question 5 (graph)

NO CALCULATOR

The graph of $f(x) = \frac{2x}{\sqrt{x^2+1}}$ is shown with its two horizontal asymptotes. What is $\lim_{x\to-\infty} f(x)$?



(A) 2
(B) 0
(C) -2
(D) $-\infty$

Solution
Answer: (C). To the left the graph levels off toward $y = -2$. Algebraically, $\sqrt{x^2} = -x$ for $x < 0$, so $f(x) \to \frac{2x}{-x} = -2$.

Question 6 (Free response)

NO CALCULATOR

Let $f(x) = \frac{4x}{\sqrt{9x^2+1}}$.

(a) Find $\lim_{x\to\infty} f(x)$.
(b) Find $\lim_{x\to-\infty} f(x)$. Show why the sign changes.
(c) Write the equations of all horizontal asymptotes of f.
(d) Find $\lim_{x\to\infty} \frac{2e^x+5}{e^x-3}$.

Solution
(a) For large $x > 0$, $\sqrt{9x^2+1} \approx 3x$, so $f(x) \to \frac{4x}{3x} = \frac{4}{3}$.
(b) For $x < 0$, $\sqrt{9x^2} = 3|x| = -3x$, so $f(x) \to \frac{4x}{-3x} = -\frac{4}{3}$.

(c) $y = \frac{4}{3}$ and $y = -\frac{4}{3}$.
(d) Divide by e^x: $\frac{2 + 5e^{-x}}{1 - 3e^{-x}} \to \frac{2}{1} = 2$.